

Material Test Report / Certificate LSM-24-18877

Lone Star Specialty Metals, Inc. | 8800 Wallisville Rd, Houston, TX 77029

Certificate No. / Date	LSM-24-18877 / 2024-09-12
Heat / Batch No.	8Q4412
Customer	American Completion Tools, Inc.
Purchase Order	ACT PO-87455
Product	Hot rolled bar, 4.50 in. dia x 20 ft R/L, quenched & tempered
Specification(s)	ASTM A29 / AISI 4140; ACT-MS-01 Rev C supplementary
Quantity	24 bars / 14,120 lb

Chemical Composition (wt %)

Element	Value	Element	Value
C	0.41	Cr	0.98
Mn	0.92	Mo	0.19
P	0.011	Ni	0.08
S	0.018	Cu	0.14
Si	0.26	Fe	Bal.

Mechanical / Physical Properties

Property	Result	Requirement
Yield strength (0.2% offset)	121,300 psi	>= 110,000 psi
Tensile strength	138,700 psi	>= 125,000 psi
Elongation in 2 in.	18.5 %	>= 15 %
Reduction of area	52 %	>= 45 %
Hardness	30 HRC	28 - 32 HRC
Charpy V-notch @ -4 degF (avg of 3)	42 ft-lb	>= 30 ft-lb

Thermal Treatment

Austenitized 1,575 degF 2 hr, oil quench; tempered 1,120 degF 3 hr, air cool

NDE / Additional Testing

UT per ASTM A388 (flat-bottom hole cal): no recordable indications

Sour Service (NACE MR0175 / ISO 15156) Statement

Not certified for NACE MR0175 direct exposure at this hardness (28-32 HRC). For sour wetted service ACT specifies L80-13Cr or ACT-MS-02 low-hardness 4140 (22 HRC max).

Remarks

Material melted and manufactured in the USA. Fully killed, fine grain practice, grain size 7 or finer per ASTM E112. Mercury-free. This certificate complies with EN 10204 Type 3.1. Used by ACT for: PKR-4500 mandrels/subs (see QIR-2025-0142), ST-10 housings, general 4140 tool bodies.

Certification

We hereby certify that the material described above was manufactured, sampled, tested and inspected in accordance with the referenced specifications and was found to meet those requirements. This certificate is issued in accordance with EN 10204 Type 3.1.

Role	Name	Signature	Date
Authorized Signatory	Q. Ramos, Chief Metallurgist, Lone Star Specialty Metals	/signed electronically/	2024-09-12
ACT Receiving QA Review	S. Nguyen	/signed electronically/	2024-09-12